

Rushil Singh

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Education

Carnegie Mellon University

Pittsburgh, PA

Bachelor of Science in Information Systems, Minor in Artificial Intelligence

Expected May 2028

Relevant Coursework: Data Structures & Algorithms, Database Design & Development, Natural Language

Processing, Designing Human-Centered Software, Discrete Math, Probability Theory, Linear Algebra, Calculus 3

Professional Experience

Perforated AI

Pittsburgh, PA

Machine Learning Research Collaborator

May 2026 – Present

- Developed PyTorch-based transformer efficiency techniques using Perforated Backpropagation and artificial dendrite architectures, reducing model parameters by 60% while accelerating trading-model inference by 15%+
- Conducted W&B hyperparameter sweeps and multi-seed experiments across 15+ parameter combinations, systematically evaluating dendrite thresholds, regularization, and module placement against baseline performance

ListenFirst

New York City, NY

AI Engineering Intern

July 2026 – August 2026

- Built AI-powered reporting workflows as the Partnerships unit's sole intern, deploying organization-wide Claude Skills using Python, JavaScript, and RESTful APIs that reduced manual effort by 75% across 60+ projects
- Developed production automation for monthly billing and scheduling across 40+ client accounts, cutting API call volume by 90%+ through streamlined data pipelines and surfacing \$15K+ in billing discrepancies missed manually

Twingly

Remote, EST

Software Engineer

June 2026 – August 2026

- Built and shipped cross-platform AI agent capabilities in Swift and Python, developing voice cloning, reservation booking, and automated gaming features accessible through natural prompts for 100+ users and 2,000+ followers
- Contributed through code reviews and PR/MR-based feature integration while deploying Electron desktop builds across macOS and Windows and resolving platform-specific issues to maintain consistently updated releases

Activities & Involvement

180 Degrees Consulting

Pittsburgh, PA

Student Consultant

Active Member

- Selected from a competitive applicant pool to complete consultant training in public speaking, client communication, and presentation design in preparation to support Pittsburgh non-profits on real-world consulting projects

Traders @ CMU

Pittsburgh, PA

Prediction Market Analyst

Active Member

- Built a cryptocurrency arbitrage tool achieving a 65% win rate by fine-tuning an ML model and backtesting strategies through trade simulations to identify discrepancies between Kalshi's BTC prediction markets and their price sources

Projects

SquidSpell – ASL Fingerspelling Tool | *Python, scikit-learn, MediaPipe, FastAPI, WebSockets, React, Supabase* | [Live Demo](#)

- Built a real-time Computer Vision app that runs browser hand tracking to recognize the ASL alphabet with MediaPipe and streams data over WebSockets to a Python service that classifies each letter at 99.4% test accuracy
- Deployed the full stack on Vercel and Render with 200+ automated tests, using Supabase (PostgreSQL, RLS, Google OAuth) for accounts, saved history, and a global speed-test leaderboard, with all game scoring handled on server-side

Snake Duel – Real-Time Multiplayer Game | *C++, WebAssembly, SDL2, WebSockets, Docker, CMake* | [Live Demo](#)

- Created a real-time online and local 1v1 multiplayer game in C++, sharing one game-logic core across a desktop app, a WebAssembly browser build, and the match server so every device remains synced identically with the server
- Shipped a 10 Hz authoritative WebSocket game server with an in-memory room registry and a JSON protocol, deployed as a single Docker container, with 73 automated tests spanning the simulation, bot AI, and wire protocol

Technical Skills

Languages: Python, Java, C, C++, JavaScript, TypeScript, Swift, SQL, HTML, CSS

Frameworks & Libraries: PyTorch, React, Node.js, Flask, Pandas, NumPy, Matplotlib, Framer Motion, Tailwind CSS

Developer Tools: Git, GitHub, Linux, Docker, GitHub Actions, Supabase, PostgreSQL, Jupyter, Claude Code